

Article

State of the Art of Digital Twins in Improving Supply Chain Resilience

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Abstract: *Background:* In today's complex and rapidly changing global markets, supply chain resilience (SCR) has become critical for businesses aiming to maintain continuity and competitive advantage. Disruptions and challenges in the supply chain will always exist; therefore, preparing in advance and improving resilience for the upcoming consequences should be the utmost important goal. *Methods:* Digital twins (DTs) provide a comprehensive view of product performance, enabling companies to adopt an end-to-end approach to product management. To maximize product and service value, supply chains must also be managed holistically. *Results:* Therefore, companies will need smarter strategies to balance inventory costs, availability and lead times. The optimal setup of suppliers, manufacturing, logistics and stock locations will ensure high service levels and meet customer expectations. Additionally, supply chains must be resilient, and capable of maintaining performance during disruptions and adapting to demand changes. *Conclusions:* This paper aims to provide a comprehensive review of the state of the art in digital twin applications within supply chains, focusing on their role in improving visibility, agility and decision-making. This paper explores how digital twins might integrate with emerging technologies such as IoT, AI and blockchain, fostering a more adaptive and robust supply chain ecosystem.

Keywords: digital twins; supply chain resilience; IoT; AI; disruption management; optimization; systematic literature review



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1. Introduction

The rapid advancement of cyber-physical systems has drawn increasing attention to the concept of digital twins and their potential to revolutionize various industries, including supply chain management [1]. By creating a comprehensive digital representation of the supply chain, organizations can better understand their operations' complex interdependencies and vulnerabilities, ultimately improving supply chain resilience. These latest technological breakthroughs have revolutionized SCM (supply chain management) processes as businesses and industries increasingly recognize technology as a powerful strategic tool for ensuring sustainable performance and future success [2]. Recent studies highlight that digital twins in supply chain management provide advantages like simulating real-time conditions, enabling intelligent decision-making and offering cost-effective, tailored solutions while facilitating team coordination and monitoring across supply chain links [3].

While the interest in digital twins and their potential in supply chain management is growing, there is still a pressing need for a comprehensive review. Only a few papers

explore how digital twins integrate with AI, IoT and blockchain technologies in order to create a cohesive, intelligent and resilient supply chain. Most studies focus on individual technologies without examining their combined impact on supply chain resilience and optimization. This article explores digital twins in the context of supply chain management and resilience. It begins by defining key concepts and highlighting the importance of resilience in adapting to disruptions while maintaining efficiency. The discussion then shifts to algorithms' evolution and application in digital twin technologies, focusing on research methodologies and significant advancements over time and across regions. This structured approach seeks to offer valuable insights into the integration of digital twins and the ongoing development of supply chain strategies in a rapidly changing global landscape. Additionally, the integration of digital twins with blockchain and AI highlights their transformative potential in fostering innovation and sustainability in supply chain management and its resilience [4].

2. Supply Chain Management and Resilience

At the root of the development of the term supply chain management (SCM) is the evolution of logistics and how its role in an organization has changed over time because of the influence of globalization. On the one hand, logistics creates the framework and the direction in which the material and information flows in a company are to be organized; on the other hand, supply chain management comes to its aid to try to create connections between the processes carried out by other entities in the supply chain, such as suppliers, organizations, and customers [5].

The objective of supply chain management is to optimize internal planning and resource allocation of a company to facilitate effective relationships with other actors in the supply chain [6]. In the literature, the relationship between logistics and SCM appears under different forms, such as logistics as an integrated part of supply chain management, SCM as part of logistics, SCM instead of logistics or logistics, and SCM as two independent disciplines.

The field of supply chain management (SCM) is characterized by volatility and novelty, which has resulted in the inability to establish a universally valid definition over time. Therefore, SCM is variously defined in literature: on the one hand, as a management approach, and, on the other hand, as a set of activities to implement a management approach [7].

In 2017, LeMay et al. proposed a more elaborate and comprehensive definition, which states that "Supply chain management is the design and coordination of a network through which organizations and individuals get, use, deliver and dispose of material goods; acquire and distribute services; and make their offerings available to markets, customers and clients" [8]. Thus, it can be stated that SCM encompasses the planning, coordination and management of all activities involved in sourcing, procurement, production and logistics. Traditionally, SCM focused heavily on efficiency—specifically, cost reduction, lean operations and inventory minimization.

Over the years, strategies such as just-in-time (JIT) production and lean manufacturing became popular, where businesses minimized inventory levels to reduce costs and operated with highly efficient but tightly interconnected supply chains. As SCM became more globalized, these practices, while efficient, also increased vulnerability to disruptions. A disruption at one point in the supply chain could have far-reaching effects on other nodes, creating substantial risks for businesses. This is where the concept of resilience becomes crucial. In the wake of increasing natural disasters, climate change and rapid urbanization, resilience has become a key focus in urban planning and infrastructure.

Considering the growing intricacy of global supply chains, particularly as a result of significant events such as the 9/11 attacks, Hurricane Katrina and the 2008 financial crisis, the notion of supply chain resilience has emerged as a significant area of concerns.

The concept of supply chain resilience is centered upon the capacity of supply chains to recuperate from unforeseen disruptions and to maintain operational efficacy [9–11].

Resilience in SCM is defined as the ability of a supply chain to anticipate, prepare for, respond to and recover from disruptions while maintaining its essential functions [12]. This concept shifts the focus from merely optimizing efficiency to building flexibility and adaptability into supply chains. In academic literature, the interaction between SCM and resilience is often framed in terms of balancing the need for efficiency with the ability to handle unexpected shocks.

In their contribution to the field, Christopher and Peck (2004) were among the first to emphasize the necessity of resilience in SCM, particularly in global supply chains. The authors suggest that conventional SCM systems, which prioritize cost reduction and JIT principles, are insufficiently flexible to cope with the contemporary risks faced by modern businesses [5]. Consequently, they propose that resilience should be regarded as an essential component of efficiency. Their findings indicate that the leaner and more efficient a supply chain is, the more vulnerable it becomes to disruptions, particularly in the absence of buffers such as inventory stock or alternative supply routes.

Ponomarov and Holcomb (2009) also elaborate on the interaction between SCM and resilience by developing a framework for understanding supply chain resilience. They argue that resilience is not just about responding to disruptions but also about anticipating risks and designing supply chains that can absorb shocks without collapsing [13]. This involves creating flexible sourcing strategies, diversifying suppliers and investing in technologies that provide real-time visibility and predictive analytics. Their framework emphasizes that resilience requires a shift in mindset from viewing SCM as a purely operational process to seeing it as a strategic tool for managing risk and ensuring long-term business sustainability.

The 2011 Tohoku earthquake and tsunami in Japan provides a notable example of the application of resilience in SCM in the aftermath of a significant disruption. The earthquake had a profound impact on global supply chains, particularly in the auto-motive and electronics sectors, given Japan's status as a major supplier of essential components such as semiconductors and automotive parts.

Research conducted by Revilla and Saenz (2014) demonstrates how companies like Toyota employed resilience strategies to facilitate recovery from this disruption. The automotive manufacturer had previously relied on lean manufacturing and just-in-time inventory practices, which left it susceptible to vulnerability when its suppliers were affected by the disaster. In the aftermath of the earthquake, Toyota implemented the "Rescue" initiative, which aimed to enhance the resilience of its supply chain. This involved diversifying suppliers, ensuring geographic redundancy and increasing transparency and communication with sub-tier suppliers. These measures enabled Toyota to recuperate more expeditiously from the disruption and minimize the impact on its global operations [14].

As global supply chains continue to face new and emerging risks, resilience will remain an essential factor in ensuring that businesses can survive and thrive in an unpredictable world. While traditional SCM focused primarily on efficiency, the increasing frequency and severity of disruptions have led to the integration of resilience as a key component of supply chain strategies. Resilience in SCM is not just about responding to crises but about designing supply chains that are flexible, adaptive and capable of withstanding disruptions.

3. Research Methodology and Framework

One of the aims of the present paper is to outline the developments and changes in SCM from the rudimentary technology that was available decades ago to the awe-inspiring digital twin technologies of today, which have transcended the boundaries of imagination and revolutionized the SCM landscape. A systematic literature review (SLR) is conducted

in order to better understand the analysis of the algorithms and techniques' evolution in SCM. The research methodology was divided into four main steps, as shown in Figure 1. Each step helps to reduce possible errors and prejudice in assessing the literature review of the present paper.

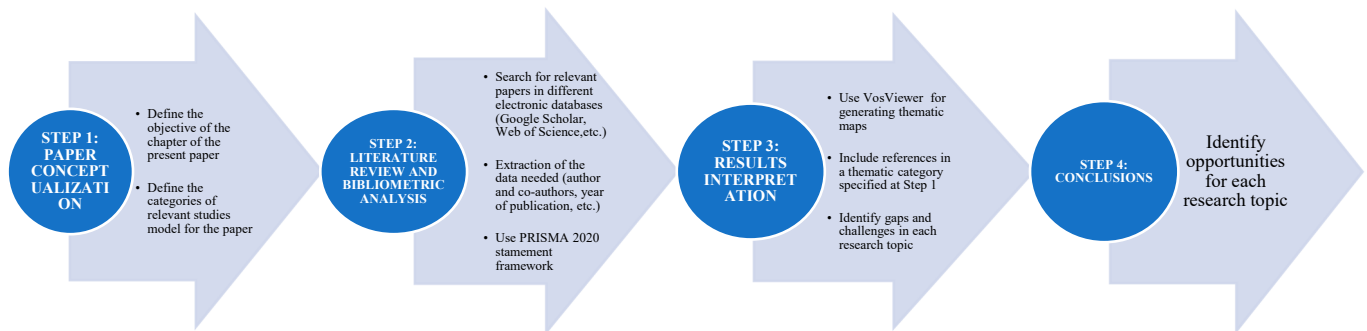


Figure 1. Phases of the literature review used for the present paper.

In the first step, we needed to both define the aims of this chapter of this paper (represented by the insights of the algorithms' evolution in supply chain over time) and structure the relevant papers we find into different categories. We divided the literature into eight main topics: early foundations of supply chain and operation research, evolution of algorithms in supply chain, the growth of enterprise resource planning (ERP) and supply chain software, and the role of internet and cloud technologies. Other topics taken into consideration are artificial intelligence, machine learning and predictive analytics in supply chain, supply chain analytics, big data, IoT and Industry 4.0, and blockchain and digital twins technologies in SCM and resilience. The latter represents two distinctive categories in our literature research.

The second phase of the literature review consisted of finding relevant papers in available electronic databases such as Google Scholar, Web of Science, etc. Furthermore, a thematic analysis was conducted to sort the papers according to each category and to gather information regarding the author, co-authors, year and country of the paper, publication in a journal, and the number of citations. The literature research was conducted based on both the topics mentioned above and the PRISMA 2020 Statement (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [15]. We opted for this approach as it not only ensures that our literature review is comprehensive, reproducible and transparent, but it also allows for a better understanding and critical appraisal of the review.

In order to apply the PRISMA framework in this step (depicted in Figure 2), each database/source/library was examined in terms of the search technique and keywords used. The databases we used in the literature review are Scopus/Elsevier, Web of Science, Science Direct, IEEE Explorer and Google Scholar. All searches ranged from years 1913, when we found one of the first publication related to SCM optimization, to 2024, and covered books, journal articles, chapters in books, review papers and research reports (all of them written in English). Using the selection criteria based on the PRISMA framework, a total of 250 articles that mainly covered the main eight topics mentioned above were found.

Our research used only original research publications, review papers, books and chapters in books. In the next step, all duplicates found were examined and removed from our systematic review in order to ensure its quality. Moreover, the papers' abstracts were analyzed to ensure the relevance of the academic material included in the review process. In the end, 223 articles were chosen for screening after evaluating them against the inclusion and exclusion criteria.

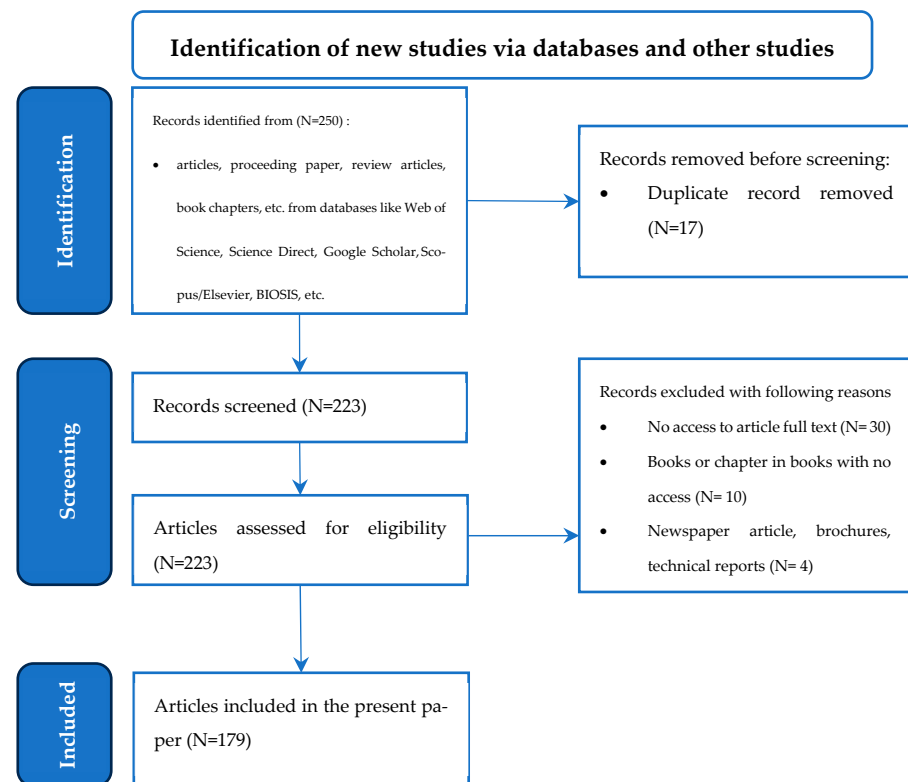


Figure 2. Application of PRISMA 2020 flowchart for the literature review in this paper.

In the exclusion phase, articles with no access to full text and those that were brochures, technical reports, newspaper articles were removed from the literature review. The analyzed papers should be written in English and meet one of the abovementioned eight topics taken into discussion. Excluded articles were published between 1952 and 2019 and were from a wide range of countries worldwide not focusing on a particular country or countries. After applying the PRISMA framework, a total of 179 articles were finally chosen for the systematic literature review.

These pieces of information allowed us to advance to the third phase of our methodology where, for a comprehensive analysis of the papers included, we opted for VOSViewer and CiteSpace to capture and analyze the appearance of these keywords from the 179 articles considered at this point.

VOSViewer is regarded as a software tool for creating and exploring maps based on network data and explores co-authorship, co-occurrence, citation, bibliographic coupling and co-citation links in one of three possible representations: network, overlay, or density visualization [16]. CiteSpace is well known for its capabilities to conduct visual analytic study of the scholarly literature of a research field, a research area or a discipline, collectively known as a knowledge domain [17].

In VOSViewer, a binary counting approach was used in order to obtain Figure 3A regarding the most frequently occurring keywords found in our literature review. It should be noted that each point in the heat map has a color assigned that represents the density of items at that specific point. Colors range from blue to red as an initial setting of VOSViewer 1.6.20 software. It must be highlighted that a point color is determined by the number of items in its vicinity and the weights of the neighboring items. If a point has not only a higher number of items but also heavier weights in its neighborhood, it will be colored red. On the other hand, if a point has a lower number of items and lighter weights, it will be colored blue. In CiteSpace software, the time slice was set to one year (starting from 1913 to 2024) and the measurement was set to intensity analysis.

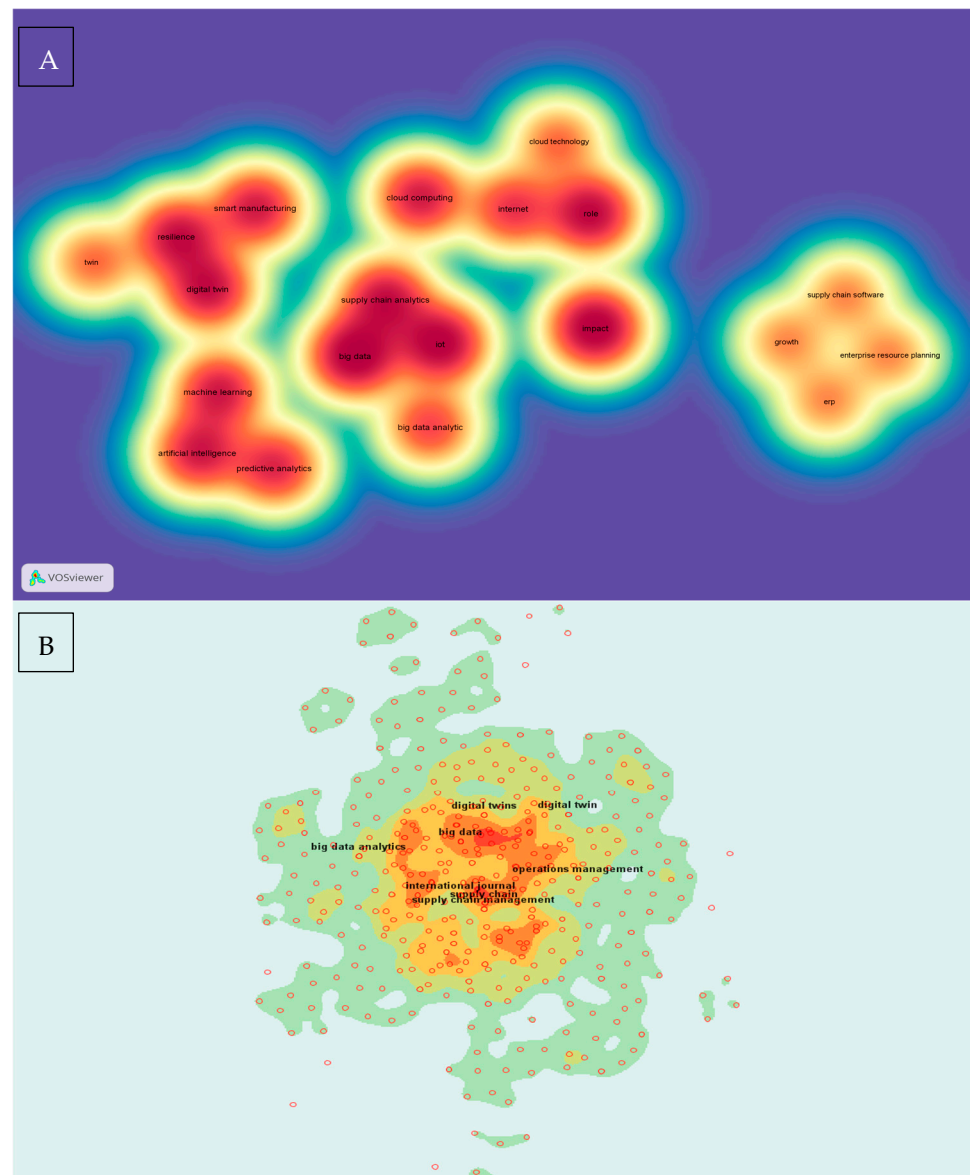


Figure 3. Heat map of word frequency from VOSViewer (A) and CiteSpace (B) used in the present systematic literature review.

From this figure, it can be observed that words like digital twin, resilience, supply chain analytics, big data and IoT have a higher frequency in our systematic literature review due to the reasons specified above.

Furthermore, a keyword clustering analysis (depicted in Figure 4) was created in order to outline the focus of emerging technologies (IoT, digital twins, AI) and supply chain. The VOSViewer map focuses on keyword co-occurrence (thus helping to identify thematic clusters their interrelationships), while the CiteSpace network aims to provide a citation-based perspective (in this way, influential clusters and research trends are found). Both maps highlight the term digital twin as a central concept as it is associated with resilience, smart manufacturing and prediction in supply chains. The CiteSpace clusters (Nos. 0 and 5) emphasize digital twins in the context of both SCM and SCR, highlighting their importance in reducing vulnerabilities and enhancing adaptability. From the VOSViewer perspective, existing connections between terms such as digital twin, resilience and smart manufacturing underline the role of digital twins in creating adaptive and highly verified manufacturing systems. Nevertheless, both visualizations show the increasing adop-

tion of digital twins and how traditional SCM is transformed by enabling predictive analytics and enhanced real-time monitoring in order to create robust and resilient supply chains.

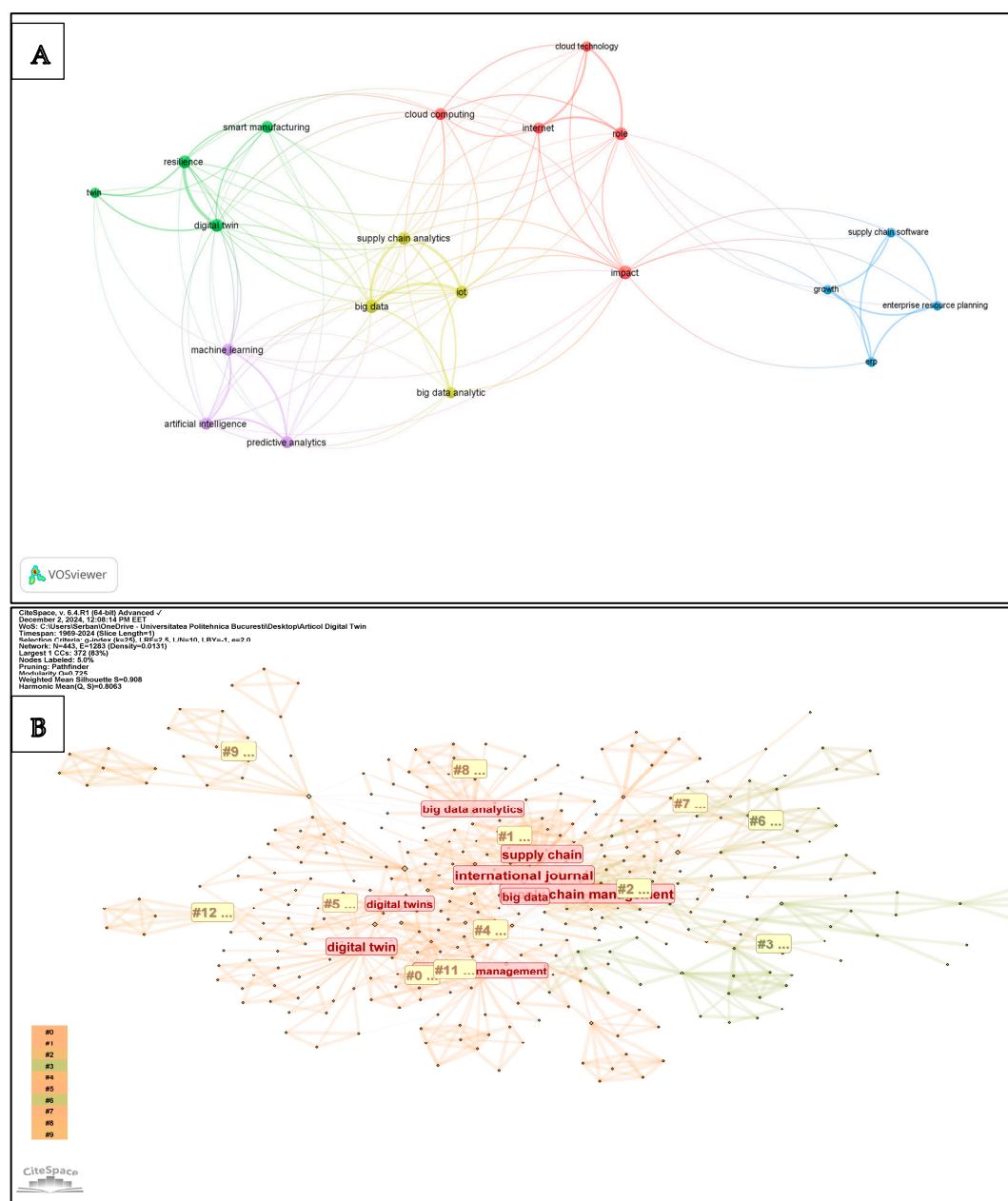


Figure 4. Analysis results of keyword clustering from VOSViewer (A) and CiteSpace (B).

The next big clusters identified are related to IoT, big data analytics, AI and predictive analysis (yellow and purple clusters in VOSViewer, clusters Nos. 4 and 1 in CiteSpace). From the map generated by VOSViewer, these can emphasize the reliance of modern supply chains on real-time and data-driven decision making, while the CiteSpace cluster No. 4 highlights a bibliometric hotspot focusing on the application of data science in supply chain optimization. Furthermore, from the strong connection between IoT and big data terms from VOSViewer map, the importance of sensor-based and real-time data collection in supply chains can be shown. As a result, more and more organizations are starting to invest in data infrastructure and IoT technologies to improve demand forecasting.

As a next step in this phase, we included each article to its specific category defined at phase one and we identified the gaps and challenges in each research topic. The final phase

consists of outlining the research opportunities identified for each topic. By following the first three of the four phases mentioned above, we finally included 179 relevant articles in our systematic literature review.

The number of articles and their percentages for each category considered are shown in Figure 5, from which we can observe that most of the articles are related to the topic of digital twins in supply chain management and resilience (50 articles, which represent 27.93%), followed by supply chain analytics, big data, IoT and Industry 4.0 (35 articles, representing 19.55% from the total), the role of internet and cloud technologies and blockchain in supply chain management (with 21 and 22, respectively, articles taken into account).

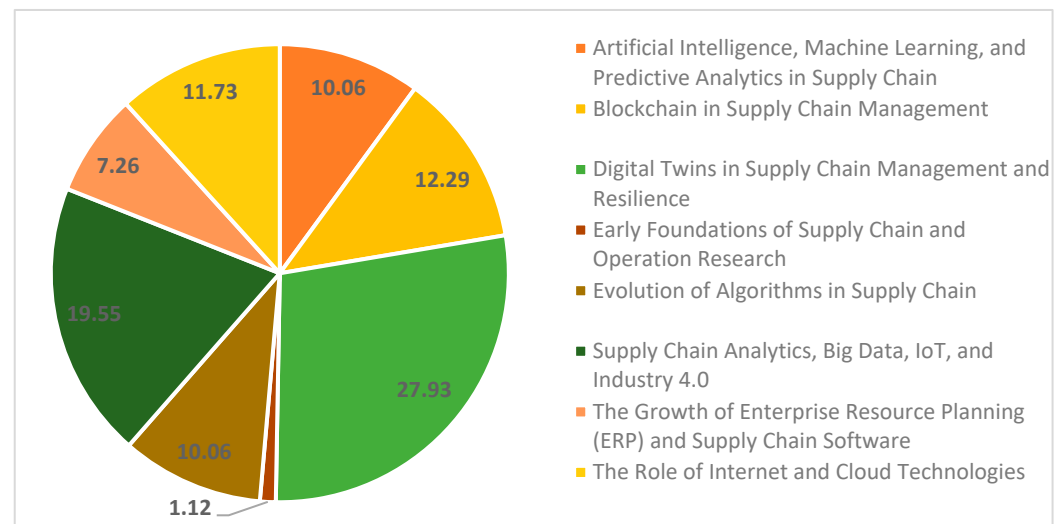


Figure 5. Number of articles per category and their percentages considered for the literature review of algorithms and techniques evolution in SCM.

The distribution in time of the papers' publications (between 1913 and early 2024) is shown in Figure 6, from which it can be observed that most of the articles appeared in 2020 (26 articles were published that year), 2021 (18 papers), and in 2019 (15 articles). The main topics debated during these years by the researchers mostly represented the effects of the implementation of both blockchain and digital twins in SCM and different solutions based on artificial intelligence, machine learning and IoT in supply chain analytics. As most of the papers investigating the mentioned topics have been appearing since 2018, it can be predicted that the number of articles in this research field will continue to grow.

Also, of great importance to our literature research on the algorithm's evolution during this period from initial and rudimentary ones to the development of digital twin technology is the tracking of the country where these papers and books originated (Figure 7). From this perspective, we can observe that in journals from the United States and the United Kingdom, 74 and 60 papers were published, respectively. Most of the topics that were considered in the relevant papers from the United States are related to algorithms' evolution in supply chain, the growth of ERP and the internet and cloud technologies' importance in supply chain, while in the United Kingdom, most of the articles analyzed blockchain application in supply chain and big data, IoT and Industry 4.0 impact. In third placed is the Netherlands, where 23 articles were published in its journals on topics such as algorithms' evolution and supply chain analytics.

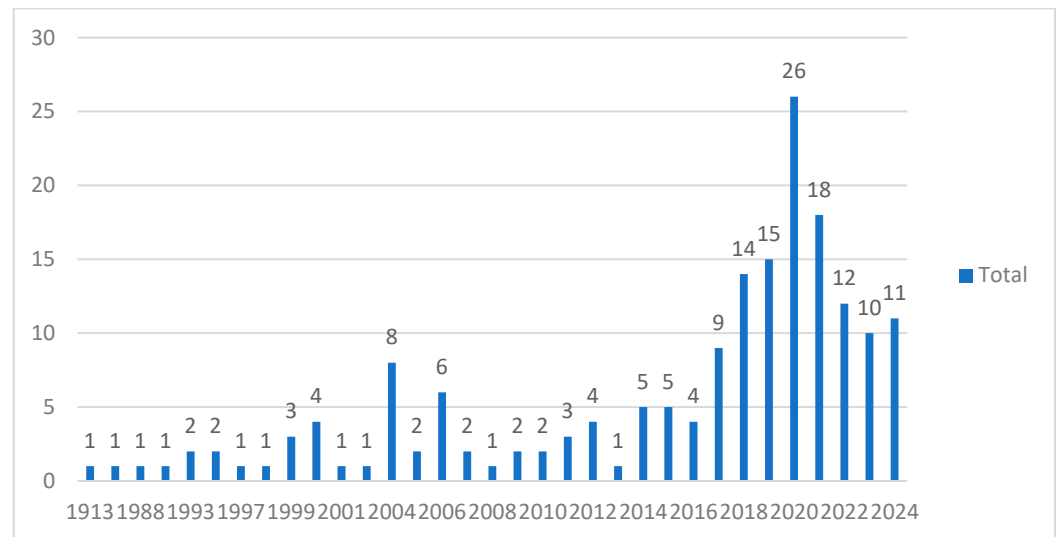


Figure 6. Number of articles published between 1913 and 2024 regarding the evolution of algorithms and techniques in SCM.

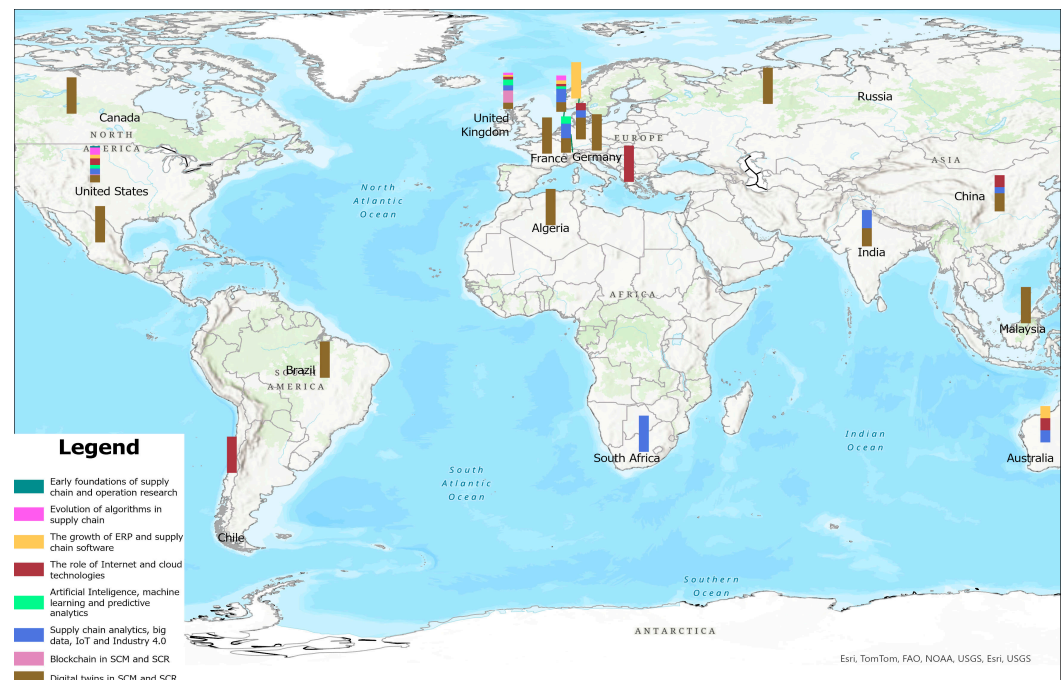


Figure 7. List of countries where the relevant papers of the conducted research mostly appeared.

As a brief conclusion of this chapter, observing the broad interest in supply chain management and resilience and the recent developments in this field, it can be predicted that the number of articles will continue to grow in the future. By integrating VOSViewer and CiteSpace into our systematic literature review, it was revealed that both SCM and SCR are moving toward interdisciplinary convergence.

Key directions include automatization and digitalization with the aid of emergent technologies (AI, ML and digital twins) and real-time data and cloud integration to create resilient, adaptable and sustainable supply chains. In the next chapter of our paper, each research topic will be discussed in order to show how it shaped the development of SCM and SCR.

4. Smart Supply Chain: Exploring the Industry 4.0 Landscape

4.1. Early Foundations and Algorithms' Evolution in Supply Chain and Operation Research

After highlighting this information regarding the number of articles published per each year and country, the present paper will briefly present those papers that impacted the SCM starting from the first inventory management models and ending with the implementation of the most recent discoveries such as blockchains and digital twins. The roots of SCM can be traced back to the early 20th century, when theories and models were developed to address the growing complexity of inventory control and production planning. The use of computers for decision-support systems in supply chains led to the development of algorithms that can address the complexity of decisions that business operations face on a daily basis. Significant developments in supply chain optimization techniques occurred from the 1980s through the early 2000s as computers led to the introduction of heuristic and metaheuristic approaches, risk-pooling models and integrated production–distribution systems. Eppen and Schrage (1981) proposed centralized ordering policies for multi-warehouse systems with lead times and random demand [18,19], while Cohen and Lee developed strategic models for integrated production–distribution systems (their work highlighted the need for synchronization between manufacturing and distribution operations and the importance of considering production capacity, transportation costs and lead times when designing supply chains) [20].

Schwarz introduced a risk-pooling model that helped companies address demand variability by centralizing inventory management across multiple warehouses. His model allowed firms to reduce safety stock levels while maintaining service levels, thereby reducing the risk of stockouts [21]. Vollmann, Berry and Whybark developed a comprehensive manufacturing planning and control system framework, which incorporated production scheduling, material requirement planning and distribution planning into a unified system [22].

Metaheuristic algorithms became increasingly important in the optimization of supply chain operations. Osman's paper from 1993 first applied simulated annealing and tabu search algorithms to the vehicle routing problem to provide near-optimal solutions to problems that were too large or complex to be solved by traditional optimization methods [23].

The use of metaheuristics extended beyond vehicle routing to include scheduling and inventory management problems. The importance of flexibility and robustness in supply chain design was further emphasized by Fisher, who developed a model for determining the optimal supply chain strategy based on the nature of the product being produced [24,25]. Fisher's work showed that supply chains for functional products with stable demand should focus on efficiency.

4.2. Enterprise Resource Planning Models

As a next step in the algorithm's evolution, the Enterprise Resource Planning (ERP) emerged as an enterprise information system designed to integrate and optimize the business processes and transactions in a corporation. According to Kurbel, two main approaches anchor the development of ERP: the first one supposes that companies carry out most of their work within business processes (involving many business functions) and the second one is represented by the need for effective information systems [26].

The ERP systems began to take the shape we know nowadays mostly in the 90s, when Christopher stated that ERP systems play a crucial role in logistics and SCM by reducing costs and improving service levels. He also considers this enables companies to manage complex supply chains more effectively by providing real-time visibility into operations, allowing for faster and more accurate decision-making [27]. Lee and Billington examined in 1993 how ERP systems enhanced material management in decentralized supply chains, where different entities within the supply chain could operate autonomously. They found

that ERP systems facilitated better coordination among supply chain partners, reducing lead times and improving inventory management [28].

Davenport noted in 1998 that ERP systems were critical for companies operating in complex and global supply chains, enabling real-time data sharing and collaboration across different locations and functions [29]. Moreover, he highlighted the risk of ERP systems becoming overly rigid, as companies might feel constrained by the system's preconfigured processes. He also stated that this rigidity could limit a company's ability to respond to market changes and customer demands. To address this challenge, some businesses opted for modular ERP systems for greater flexibility and customization [29].

ERP systems have long served as an important element of supply chain integration and process optimization. However, their rigid architecture can sometimes hinder flexibility, particularly when companies need to respond swiftly to unexpected market dynamics [30]. While the advent of cloud-based ERP systems has alleviated some of these concerns, challenges related to data security and the dependence on stable internet connectivity remain significant hurdles.

4.3. Role of Internet and Cloud Technologies in SCM and SCR

The early 2000s saw the emergence of cloud-based ERP solutions, which further transformed supply chains. Cloud-based ERP systems offer businesses a more cost-effective and scalable solution. In 2002, Gattiker and Goodhue conducted an empirical study on the impact of ERP-driven changes on business processes and found that ERP implementation often required substantial organizational changes. Businesses had to adapt to new workflows, retrain employees and invest in IT infrastructure to support the system [31]. In 2003, Hugos highlighted the importance of cloud technologies in enabling supply chain agility and responsiveness. He considers that ERP cloud-based systems allow companies to access real-time data from anywhere globally, providing greater flexibility in managing global supply chains [32].

The integration of IoT and cloud-based ERP systems has revolutionized supply chain operations by enabling improved connectivity, scalability and responsiveness. On the other hand, these advancements come with their own set of challenges. The use of IoT in supply chains generates big volumes of data, which can overwhelm existing data processing capabilities, which results in scalability issues [33]. Furthermore, the lack of standardization in IoT protocols results in interoperability challenges, while the continuous power requirements of IoT devices pose obstacles, especially in underdeveloped regions [34].

Other scientific articles, books or chapters in books related to this topic and with great impact on SCM and resilience (mentioning whether these studies are empirical or conceptual ones and what their scope is) are presented in Table 1.

It can be stated that the rapid advancement of internet and cloud technologies has revolutionized SCM by enabling companies to achieve excellent connectivity, scalability and flexibility. These technologies facilitate real-time data sharing, seamless collaboration and enhanced visibility across global supply chains, enabling organizations to respond more effectively to market demands and disruptions.

One of the main benefits of the internet and cloud technologies is the enhanced visibility they provide across the entire supply chain. Integrating cloud technologies with supply chain operations allows businesses to monitor the movement of goods in real-time, track shipments and manage inventory levels more effectively. This visibility enables companies to reduce lead times, minimize stockouts and improve overall supply chain performance [23].

Table 1. List of scientific research that outlines the impact of internet and cloud technologies on SCM and SCR.

Category	Reference	Research Objective	Research Type
The role of internet and cloud technologies	Buya et al. (2009) [35]	Analyzes cloud computing as a transformative technology that enables computing to be delivered as a utility (like water or electricity). The study discusses market-oriented resource allocation, virtualization and data center management to support scalable and efficient cloud infrastructures. Identifies and analyzes the bullwhip effect in supply chains, which arises from information distortion. It also proposes strategies for mitigating its impact. The study emphasizes how better information integration can improve production planning and inventory management. Evaluates the impact of IT on supply chain capabilities and firm performance using a resource-based view. It assesses how IT adoption enhances operational efficiencies and drives competitive advantages.	Conceptual
	Lee et al. (1997) [3]	Provides a structured review of the role of information systems in supply chain management and integration. The research paper also develops a framework for understanding how IT systems can enhance supply chain efficiency and coordination. This paper examines how internet technologies use benefits suppliers in supply chain relationships, focusing on inter-organizational systems, EDI (electronic data interchange) and supply chain management systems. It also investigates how these systems contribute to intangible asset development and exploitation. The article proposes a reservoir model for federated cloud computing, which enables seamless service integration across multiple cloud platforms. It discusses service-level agreements (SLAs) and business service management frameworks to ensure reliability and efficiency. Examine how emerging e-business technologies impact organizational collaboration and performance.	Conceptual
	Wu (2006) [36]		Empirical
	Gunasekaran & Ngai (2004) [37]		Conceptual
	Subramani (2004) [38]		Empirical
	Rochwerger et al. (2009) [39]		Conceptual
	Sanders (2007) [40]		Empirical

4.4. Industry 4.0, Machine Learning and Artificial Intelligence in SCM and SCR

In the last two decades, artificial intelligence (AI), machine learning (ML) and predictive analytics have reshaped SCM and SCR. The shift from traditional, reactive supply chain operations to predictive and even prescriptive models has enabled businesses to optimize logistics, enhance inventory management and better respond to market fluctuations. One of AI's most significant applications in SCM is demand forecasting.

Traditional forecasting models rely on historical data and assumptions, which can be unreliable in volatile markets. AI systems, however, can analyze vast amounts of data, including external factors such as weather patterns, economic indicators and social media trends. Some papers highlighted the role of information systems, including AI, in supply chain integration and management. AI-driven procurement systems analyze

historical purchasing data, market conditions and supplier performance metrics to optimize procurement processes and improve supplier relationships [37].

Besides procurement optimization, integrating AI with digital twins is important to enhance supply chain operations. AI-powered digital twins enable real-time monitoring, predictive analysis and scenario simulation by leveraging data from IoT devices and historical supply chain metrics. This allows supply chain managers to anticipate disruptions, optimize logistics, and create and analyze data-driven decisions. As an example, digital twins integrated with AI technology can simulate supplier delays and automatically identify alternative sourcing strategies.

Industry 4.0, as another improvement in SCM evolution, represents the fourth industrial revolution and can be characterized by integrating digital technologies such as AI and IoT (Internet of Things) into manufacturing and supply chain operations. Its impact on SCM cannot be denied, highlighting that the digitalization of supply chains enables businesses to achieve greater automation, flexibility and responsiveness [41]. Big data analytics has become a critical tool in modern SCM by allowing businesses to process and analyze large volumes of data from different sources. It explores the role of big data analytics in logistics and SCM, emphasizing how data-driven insights allow businesses to optimize transportation routes, forecast demand more accurately and reduce operational costs [42].

Moreover, as an addition to big data analytics, the progress in neural network architectures and machine learning algorithms has greatly increased the capabilities of digital twins in supply chain management and its resilience. The most impactful technologies among them are convolutional neural networks and recurrent neural networks, transformer models and reinforcement learning. Each of these methodologies has its unique benefits, such as predictive analytics, real-time optimization and scenario simulation.

IoT enables real-time data collection and monitoring across the supply chain. IoT devices (such as sensors, RFID tags and GPS trackers) provide valuable data on consignments' position, condition and status throughout the entire supply chain. Moreover, IoT devices could improve supply chain visibility, allowing companies to track shipments in real time and respond to potential disruptions before they escalate. Some authors further explored the integration of IoT in SCM, emphasizing its role in enhancing connectivity and communication between supply chain partners. IoT devices provide continuous data streams that can be analyzed in real time, enabling businesses to optimize inventory levels, reduce lead times and improve supply chain efficiency [34].

The development of technologies such as AI, ML and predictive analytics has propelled SCM into a new era of proactive data-driven strategies. However, alongside their transformative potential, these technologies present notable challenges. AI and ML, for example, rely on vast amounts of high-quality data for training. The existence of gaps or biases in these data can compromise the accuracy of predictions that leads to suboptimal decision-making [24]. Beside these facts, the so-called "black box" nature of many AI models (particularly deep learning algorithms) complicates their explainability. This makes AI models difficult for decision-makers to fully trust their outputs. Additionally, the complexity and expertise required for implementing AI further hinders its broad adoption [43].

Other papers that best describe the impact of Industry 4.0 and the breakthrough of emergent technologies such as ML and AI in SCM and SCR are shown in the table below (Table 2).

All studies mentioned above highlight the capacity of AI and ML in revolutionizing the supply chain management landscape and its resilience overall. The two discussion topics aim to create more efficient, adaptive and resilient supply chains by integrating cutting-edge technologies with data-driven strategies. The integration of Industry 4.0 technologies has significantly improved supply chain visibility, traceability and responsiveness.

Table 2. List of relevant scientific research studies with a great impact of Industry 4.0, machine learning and artificial intelligence in SCM and SCR.

Category	Reference	Research Objective	Research Type
Artificial Intelligence, Machine Learning and Predictive Analytics in Supply Chain	Carboneau et al. (2008) [44]	Their research investigates the application of machine learning techniques (such as neural networks and support vector machines) to improve demand forecasting in supply chains. The study also aims to assess how these methods can mitigate the bullwhip effect, a common issue in supply chain management, by identifying patterns and anomalies in data that traditional models could not capture.	Empirical
	Min, H. (2010) [43]	This paper conducts a comprehensive literature review to explore the theoretical foundations and practical applications of artificial intelligence in supply chain management. The paper delves into knowledge management, decision support systems and the integration of AI in logistics, offering insights into its transformative potential in the field.	Conceptual
	Cui & Hetz (2011) [30]	In their research paper, they examine how logistics companies build and leverage networks and capabilities to enhance their service supply chain interactions. The study aims to identify the characteristics that enable firms to improve efficiency and maintain competitive advantage in dynamic logistics environments.	Empirical
	Wamba et al. (2020) [45]	The paper explores the impact of big data analytics on supply chain management practices, focusing on agility, adaptability and ambidexterity. The study aims to link big data-driven dynamic capabilities to organizational performance, offering a framework for leveraging analytics in competitive markets.	Empirical
	Zhang et al. (2023) [46]	The paper introduces a big data analytics framework tailored to cleaner manufacturing and maintenance processes. The study addresses sustainable production by incorporating advanced analytics techniques, such as data mining, to improve product lifecycle management and operational efficiency.	Conceptual
	Bayrannis et al. (2019) [47]	Their research presents a state-of-the-art review of supply chain risk management strategies using artificial intelligence. The paper identifies key research gaps and future directions, emphasizing the role of AI in proactive decision-making and mitigating disruptions.	Conceptual
	Feizabadi (2020) [24]	In his paper, the author analyzes the role of machine learning technologies in demand forecasting and their impact on supply chain performance. Furthermore, the study aims to identify methods to improve forecast accuracy and efficiency, contributing to optimized supply chain operations.	Empirical
	Bayrannis et al. (2019) [48]	The paper reviews the evolution of manufacturing solutions based on AI, focusing on sustainability. The study also explores the potential of advanced manufacturing technologies and data analytics to improve production efficiency and achieve sustainable development goals.	Empirical

Table 2. Cont.

Category	Reference	Research Objective	Research Type
Supply Chain Analytics, Big Data and Industry 4.0	Esmaelian et al. (2016) [49]	Similar to the above paper, it highlights the evolution of AI-driven manufacturing solutions with an emphasis on sustainability. The study explores the potential of advanced manufacturing technologies and data analytics to improve production efficiency and achieve sustainable development goals.	Empirical
	Toorajipour et al. (2021) [50]	The authors conduct a systematic literature review in order to explore the applications of artificial intelligence in supply chain management. The study aims to synthesize existing research and identify trends, challenges and opportunities in the integration of emergent AI technologies in logistics and operations.	Conceptual
	Miorandi et al. (2012) [33]	This paper explores the applications and challenges associated with the IoT in supply chains. It emphasizes critical issues such as interoperability, security and the integration of technologies like RFID, sensors and actuators. Moreover, it investigates how these technologies create a smarter and more efficient supply chain ecosystem while addressing challenges in adoption and implementation.	Conceptual
	Waller et al. (2013) [51]	This research analyzes the transformative potential of big data, predictive analytics and data science in redesigning supply chains. It highlights how these technologies improve collaboration, integration and education in logistics and supply chain design.	Conceptual
	Ben-Daya et al. (2019) [34]	The paper presents a literature review on IoT applications in smart supply chain processes under Industry 4.0. It focuses on supply chain integration and the potential of IoT technologies to enhance operational efficiency.	Conceptual
	Ivanov et al. (2019) [52]	The research paper aims to investigate how disruptions in a part of the supply chain propagate across the network, amplifying risks and operational inefficiencies (the ripple effect). It proposes different conceptual frameworks and analytical models to better understand and predict supply chain risks in the context of Industry 4.0. Furthermore, it aims to guide practitioners and researchers on leveraging Industry 4.0 technologies to enhance supply chain robustness and reduce vulnerability to disruptions.	Conceptual
	Müller et al. (2018) [53]	The aim of the paper is to examine how small and medium enterprises innovate their business models under Industry 4.0. Through multiple case studies, it investigates the adoption of advanced technologies to maintain competitiveness.	Empirical
	Li et al. (2018) [54]	This research paper provides an overview of the integration between 5G networks and IoT. By enabling faster data transmission, lower latency and the capacity to connect to more devices, 5G-enhanced IoT can significantly improve supply chain operations. This enhancement leads to better real-time vehicle tracking, more efficient inventory management and quicker response to disruptions, thereby bolstering overall supply chain resilience.	Conceptual

Nevertheless, the key role of big data analytics is of great importance in improving resilience by providing advanced forecasting capabilities. In this way, it enables organizations to anticipate and adapt to different kinds of uncertainties (e.g., demand volatility, supplier issues and external disruptions like the COVID-19 pandemic).

4.5. Blockchain and Digital Twin Technologies' Impact on Supply Chain Resilience

Another technology with impact on SCM and SCR is blockchain. This technology is a distributed database of records or shared public/private ledgers of all digital events that have been executed and shared among blockchain-participating agents [55]. Its history can be traced to distributed ledger technology and differs from most existing information systems designs by including four key characteristics [55]: non-localization (decentralization), security, auditability and smart execution. In blockchain, an agent creates a new transaction to be added to the blockchain.

One of the most significant benefits of blockchain in SCM and SCR is its ability to enhance transparency and traceability. Traditional supply chains rely on systems vulnerable to manipulation, errors and delays. Using blockchain, all participants can view a shared version of the truth, reducing the risk of discrepancies. A case study conducted in 2019 on using blockchain for tracking materials in decentralized supply chains found that blockchain-enabled tracking significantly improved traceability and reduced disputes [56]. Blockchain technology benefits industries where product authenticity is crucial, such as pharmaceuticals, luxury goods and food supply chains. There are articles that discussed the role of blockchain in reducing counterfeiting by creating an immutable digital record of every product's journey through the supply chain [57].

While blockchain technology offers substantial benefits, it also faces notable drawbacks that require careful consideration. Blockchain often struggles with high energy consumption, especially when utilizing proof-of-work consensus mechanisms, raising concerns about environmental sustainability [58]. Additionally, issues such as scalability and stakeholder resistance can impede its adoption across industries.

The last technology that impacted the SCM is the digital twin, which is a virtual representation of a physical object, process or system; it enables real-time monitoring, analysis and optimization of supply chain operations. This concept deals with the creation of a virtual counterpart that faithfully represents its physical counterpart in real time, thus enabling seamless convergence between the two realms [59]. It was first presented in October 2002 at a conference held in Troy, Michigan [60]. By creating a virtual model of the entire supply chain, businesses could simulate various scenarios, monitor real-time data and optimize performance. The integration of digital twins into intelligent manufacturing systems was then introduced, demonstrating how this technology could be applied to supply chain operations. It also highlighted the benefits of using digital twins for real-time monitoring and predictive analytics, which allowed businesses to optimize both their production and distribution processes [61].

Digital twins leverage real-time data from IoT devices that are combined with AI-driven analytics in order to create virtual representations of supply chain components. These virtual models enable companies to predict potential disruptions, simulate operational scenarios and enhance real-time visibility [62]. IoT and AI technologies play a crucial role in ensuring digital twins remain dynamic and adaptive, enabling resilience in supply chains [25].

The feedback loops between real-time data collection, AI processing and digital twin simulations allow for continuous improvement and increase the accuracy in decision-making. Figure 8 illustrates the integration of IoT, AI and digital twins in enhancing supply chain resilience. It highlights the data flow, actionable insights and feedback loops among

these components, showcasing their collective impact on risk management and adaptive strategies.

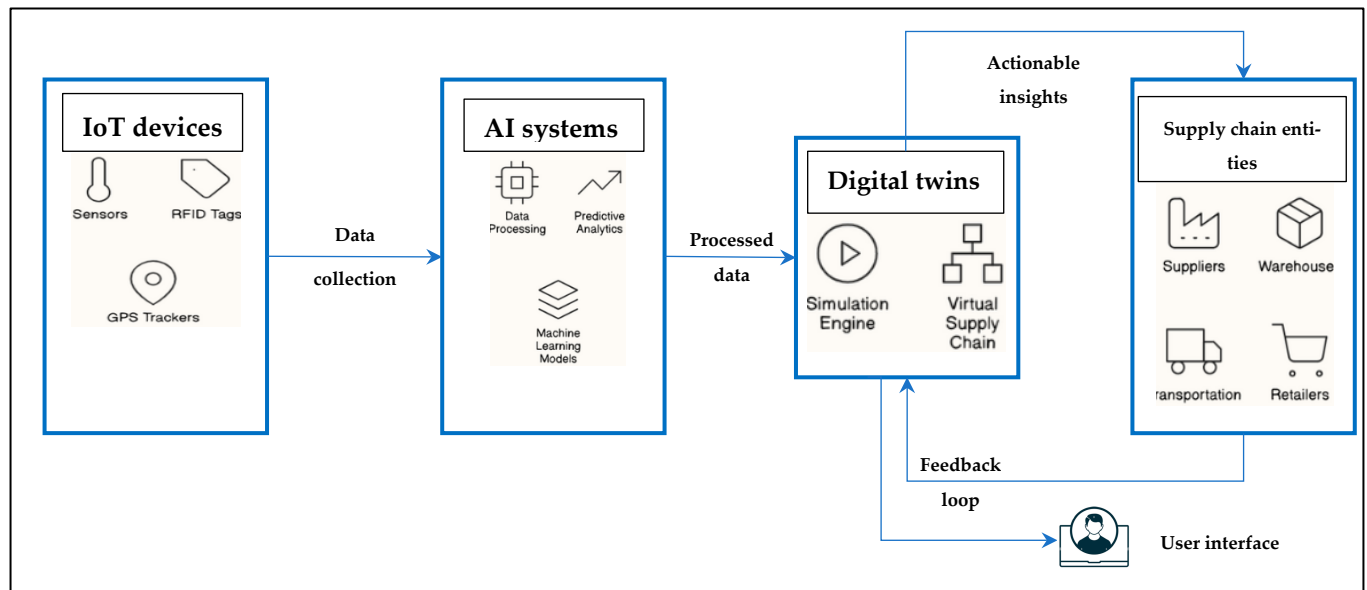


Figure 8. IoT and AI implications in digital twin applications in SCM and SCR.

Resilience in supply chains depends more and more on these technologies because they provide the capability to respond to unforeseen disruptions quickly. By integrating predictive analytics, real-time monitoring and scenario simulation, digital twins enable supply chains to transition from reactive to proactive management strategies. Moreover, adaptive strategies derived from these technologies contribute to long-term sustainability and competitiveness. Digital twin models can be modeled to manage supply chain complexities in dynamic environments (especially in emerging markets) by incorporating enhanced data integration (from sources such as IoT devices, market trend and socio-economic indicators), the use of AI-driven scenario simulation to help companies to predict outcomes and identify the right strategy, real-time adaptability to changes in the supply chain and the incorporation of regional economic data, cultural preferences and specific market risks.

AI-driven predictive analytics and scenario simulation play a key role in improving disruption forecasts and enabling digital twins to make real-time decisions. AI algorithms can analyze both historical and real-time data to identify patterns and predict potential disruptions (these forecasts enable businesses to address risks before they escalate). Predictive analytics outputs can be inserted into digital twin models, allowing them to simulate and evaluate potential responses to disruptions, while the combination between AI-driven insights and digital twins allows businesses to make informed real-time decisions.

As depicted in Figure 8, both IoT and AI serve as foundational technologies that enable digital twins to deliver predictive insights and real-time simulations. By leveraging these technologies, supply chains can achieve greater visibility, optimize operations and maintain resilience against disruptions. The iterative feedback loops shown in the diagram emphasize the importance of continuous learning and adaptation in dynamic supply chain environments.

But this heavily relies on digital twins on IoT devices, and real-time data integration comes with some disadvantages. Firstly, there are significant costs related to infrastructure, software and skilled personnel. Furthermore, the dependence on IoT devices makes these systems susceptible to cybersecurity risks and data integrity challenges. These barriers can particularly hinder adoption in smaller, resource-constrained organizations, limiting their widespread application.

Despite digital twins' capabilities to enhance supply chain management and crisis response through improved situational awareness, predictive analytics and decision-making tools, their ability to predict "black swan" events (rare, high-impact and unpredictable occurrences) remains limited. While digital twins excel in simulating known scenarios and improving system resilience, their reliance on historical and real-time data is inherently constrained by their capacity to anticipate unprecedented events. Some papers [63] identified a critical challenge in uncertainty quantification in digital twin models, limiting their predictive abilities in unforeseen events, while others [64] showed that interoperability barriers, such as the "silo effect" and fragmented data across systems, hinder digital twins from effectively identifying precursors to such events.

Despite these limitations, digital twins remain essential for fostering system resilience. Techniques such as chaos engineering can bolster their robustness against unpredictable scenarios [12]. The integration of digital twins with AI, IoT and chaos engineering helps organizations adapt to and recover from disruptive events. Future research should aim to close gaps in evidence-based assessments and adopt interdisciplinary strategies to advance DT applications in risk management and resilience-building.

A list of several relevant articles that outline the recent development of blockchain and digital twin technologies in SCM and SCR are presented in Table 3.

Table 3. List of relevant scientific research regarding blockchain and digital twin technologies impact on SCM and SCR.

Category	Reference	Research Objective	Research Type
Blockchain technology impact on supply chain resilience	Azzi et al. (2019) [65]	The authors' research analyzes the impact of blockchain on enhancing traceability and decentralization in supply chains. The study explores blockchain's ability to ensure data integrity, to streamline workflows and create a more transparent and secure supply chain ecosystem.	Conceptual
	Min (2019) [17]	The author's paper investigates the potential of blockchain technology to improve supply chain resilience. The study emphasizes how blockchain-based cryptocurrency applications and advanced architectures can lessen risks, improve agility and maintain operational continuity during disruptions.	Conceptual
	Dolgui et al. (2020) [18]	The authors of the paper present and propose a blockchain-oriented dynamic scheduling framework for supply chains. The study focuses on real-time adjustments to supply chain processes, optimizing design and mitigating disruptions (with an effect on the resilience of supply chain operations) through blockchain-powered smart contracts.	Conceptual
	Di Vaio and Varriale (2020) [66]	This study analyzes the impact of blockchain technology on sustainable supply chain performance. It uses evidence from the airport industry. The study evaluates how blockchain can improve operations management, enhance transparency and promote sustainability within complex logistical systems.	Empirical

Table 3. Cont.

Category	Reference	Research Objective	Research Type
	Saberi et al. (2019) [55]	The paper explores the role of blockchain technology in achieving sustainability within supply chain management. The study focuses on identifying key barriers to blockchain adoption represented by technological, organizational and environmental challenges. Furthermore, it examines how blockchain can support sustainable practices by enhancing transparency, traceability and accountability.	Conceptual
	Chowdhury et al. (2023) [67]	This study outlines blockchain's role in managing supply chain risks in the United Kingdom by using structural equation modeling. Also, this research focuses on how blockchain operates in VUCA environments to enhance decision-making and resilience.	Empirical
	Sheel and Nath (2019) [68]	In this study, it is examined how blockchain adoption enhances supply chain adaptability, agility and performance. The study focuses on how blockchain provides competitive advantages by improving responsiveness, alignment and overall operational excellence.	Conceptual
	Cole et al. (2019) [69]	In this research paper, the authors analyze blockchain's implications for operations and supply chain management. The paper evaluates how blockchain can improve operational efficiency, transparency and collaboration. It also develops a research agenda that highlights the strategic and operational opportunities blockchain presents for modernizing supply chain processes.	Conceptual
	Kouhizadeh et al. (2021) [58]	This study aims to investigate barriers to blockchain adoption in sustainable supply chains using a DEMATEL-based framework. The study addresses critical factors such as technology readiness, organizational culture and environmental compatibility that hinder blockchain implementation. It provides theoretical insights into the interactions between these barriers and offers practical solutions to overcome them. Blockchain's potential for efficiency, transparency and traceability is key to making supply chains more adaptive and robust, thus aligning it with resilience principles.	Conceptual
	Vu et al. (2023) [70]	This research paper examines the integration of blockchain technology within food supply chains. By enhancing transparency, traceability and data integrity (as key aspects of supply chain resilience), blockchain can improve the food supply chain's ability to anticipate, respond and recover from disruptions, thereby bolstering overall resilience.	Conceptual
	Ivanov and Dolgui (2020) [71]	The research paper proposes and evaluates the novel concept of a digital supply chain twin as a framework for managing disruption risks. It seeks to enhance supply chain resilience in the context of Industry 4.0 by leveraging advanced technologies such as digital twins, data analytics and automation. Moreover, the study explores how the digital twin concept can simulate, monitor and optimize supply chain operations in order to reduce vulnerabilities and respond effectively to disruptions, ensuring continuity and efficiency in modern and resilient supply chain networks.	Conceptual

Table 3. *Cont.*

Category	Reference	Research Objective	Research Type
Digital twins impact on supply chain resilience	Zouari et al. (2021) [72]	This paper investigates the relationship between supply chain digitalization and SCR. The study examines how the degree of digital maturity and the adoption of supply chain digital tools impact SCR. The authors employ factor analysis and structural equation modeling to assess the effects of digital maturity and digital tool adoption on SCR. The findings show that a higher degree of digital maturity significantly influences the adoption of digital tools. Moreover, as organizations advance in their digital transformation and integrate digital tools into their supply chains, they enhance their ability to anticipate, to respond to and to recover from disruptions. The importance of continuous digitalization of supply chains is outlined in order to achieve greater resilience in an uncertain environment. One of the aims of this paper is to present and explain the foundations of digital twin concept, tracing their historical evolution and examining their current state of development. It investigates the technological ecosystem that supports the implementation of digital twins (like integration of IoT components, AI and other advanced simulation technologies). Furthermore, the study also examines how digital twins contribute to the improvement in operational efficiency, optimize maintenance strategies and enables better decision-making through predictive and real-time analysis.	Empirical
	Fuller et al. (2020) [60]	The authors of this paper investigate how digital twins enhance the functionality of cybernetic and physical systems by providing real-time monitoring, control and optimization capabilities. The research examines how digital twins' integration in cyber-physical systems (e.g., supply chains) and improves system efficiency, flexibility and decision-making by enabling real-time data analysis and simulation.	Conceptual
	Negri et al. (2017) [73]	In this paper, the authors wanted to study how digital twin technology can enhance supply chain resilience during disruptions such as the COVID-19 pandemic. As an example, the disruptions in the supply chain of N95 medical masks were taken into account to illustrate how digital twins can help in mitigating such interruptions. The authors examine the impact of disruptions on supply chain performance and propose a digital twin framework to manage inventory and cash flows during such events.	Conceptual
	Lv et al. (2022) [74]	Also, the paper shows how digital twin technology can mitigate the effects of physical and financial disruptions. By integrating machine learning and simulation, the proposed framework aims to identify optimal inventory and cash replenishment policies, thereby enhancing the supply chain's ability to withstand and recover from disruptions.	Empirical
	Badakhshan and Ball (2023) [75]		

Table 3. Cont.

Category	Reference	Research Objective	Research Type
	Ivanov (2024) [76]	This research presents a framework for understanding digital twins in the context of supply chain and operations management. By integrating elements such as technology, people, management and organization, the proposed framework facilitates continuous human–artificial intelligence interactions. This leads to more reconfigurable and viable supply chains. Furthermore, this integration enhances the supply chain’s ability to adapt to disruptions and maintain operational continuity, thereby contributing in this manner to the overall resilience of the supply chain systems.	Conceptual
	Singh et al. (2023) [77]	The aim of this research is to explore how digital twin technology can be leveraged to enhance resilience and sustainability within food supply chains. Using the grey causal modeling approach, the study seeks to identify and quantify the causal relationships between critical factors influencing supply chain resilience and sustainability. By employing advanced modeling techniques, the research aims to provide actionable insights for decision-makers, enabling them to implement digital solutions that improve supply chain efficiency, reduce vulnerabilities and promote sustainable practices across the food supply chain. Another discussion topic is how digital twins contribute to reconfigurable and viable supply chains, aligning with the goals of Industry 5.0	Empirical
	Barykin et al. (2020) [78]	This research paper explores the development of a digital twin model specifically tailored for supply chains. The authors propose a digital twin framework aimed at enhancing the reliability and stability of supply chains. By integrating simulation modeling, optimization and data analytics, the digital twin model seeks to mitigate operational and critical risks, thereby improving the supply chain’s ability to withstand and recover from disruptions. In this paper, the creation and implementation of digital twin technology within supply chains is taken into account, focusing on a pharmaceutical company as a case study.	Conceptual
	Marmolejo-Saucedo (2020) [79]	The study analyzes how digital twins can improve supply chain resilience by providing real-time monitoring and control of operations across all links of the system. The author considers that by creating virtual replicas of supply chain processes, digital twins enable organizations to anticipate disruptions, optimize performance and make informed decisions, thereby improving the robustness and adaptability of business processes.	Empirical
	Ivanov et al. (2021) [80]	The paper explores digital twins, different tactical planning and resilience strategies in supply chains at a global level. It provides a decision-oriented framework for creating value through enhanced collaboration and quantitative methods.	Conceptual
	Ivanov (2022) [81]	This study was conducted to provide a theoretical framework and model integrating key dimensions of supply chain viability (agility, resilience and sustainability) while drawing lessons from the challenges faced during the COVID-19 pandemic.	Conceptual

Table 3. *Cont.*

Category	Reference	Research Objective	Research Type
	Chaouni Benabdellah et al. (2023) [82]	The objective of this paper is to address the challenges of resilient and sustainable SCM by proposing innovative approaches that align with circular economic principles. The study introduces a circular digital supply chain operations reference model to quantify the effectiveness of supply chain processes in the context of digital transformation and circular economy practices. By providing a roadmap for practitioners and managers, the paper aims to advance managerial perspectives on designing supply chains that are not only resilient and sustainable but also optimized for the technological advancements shaping Industry 4.0.	Conceptual

It can be stated that the papers above examined the potential of both blockchain and digital twins for supply chain optimization, thus emphasizing the technologies' ability to provide real-time insights into the performance of supply chain operations. Some of the studies highlighted the integration of blockchain with smart contracts, distributed ledgers and digital twins to enhance operational efficiency and trust. The potential of blockchain in fostering sustainability, particularly in green supply chain management and circular economies, was frequently emphasized. The limited outputs of these systems can also render them incapable of adapting to unexpected disruptions or rapidly changing market conditions. Supply chains have a variety of stakeholders, fluctuating demand and external uncertainties such as geopolitical risks or natural disasters. Systems with limited predictions or management options may struggle to provide the agility and foresight needed to respond effectively to these challenges. Moreover, they do not maintain consistent compatibility, with emerging technologies such as digital twins, IoT and blockchain decreasing their effectiveness significantly. Unless fully integrated, the systems would be unable to use real-time data, predictive simulations, or decisions on scale.

Therefore, these systems are important both in the assessment process for dealing with their inadequacies and exploring opportunities to make them more functional and expand their outputs to harmoniously integrate into comprehensive supply chain management frameworks. Through this, AI-based systems can provide much more holistic and adaptive solutions aligned with the increasing complexity and demands of global supply chains. The table below (Table 4) provides a structured overview of widely used AI-based systems, highlighting their features, practical applications, compatibility with digital twins and performance metrics.

Digital twins also allow for more accurate forecasting and scenario analysis, as explored in studies on the role of digital twins in managing supply chain risks, particularly in the context of disruptions caused by the COVID-19 pandemic. These studies showed how digital twins could help businesses simulate various disruption scenarios and develop more resilient supply chain strategies [81]. As mentioned above, several studies take into consideration the convergence of blockchain and digital twin technologies in order to enhance traceability, secure data sharing, and decentralize decision-making. This combination is regarded as a driver for resilience in logistics and supply chain systems, particularly in volatile and uncertain environments.

The implementation of digital twins involves significant upfront costs, particularly in adopting advanced technologies. However, justifying the return on investment (ROI) extends beyond immediate cost reductions. Enhanced visibility and reliability foster customer trust and loyalty, while sustainability benefits, such as optimized resource use and

reduced environmental impact, align with regulatory requirements and support long-term profitability. By tailoring digital twin adoption to specific business needs, conducting detailed cost-benefit analyses, and ensuring compliance with local regulations, organizations can effectively demonstrate the value of digital twins. Although initial implementation may be costly, the ability to build resilience, efficiency, and strategic growth solidifies the case for their adoption, even in low-margin supply chains [64].

Table 4. List of main AI-based digital twin technologies and their impact on SCM and SCR.

System Features	Practical Applications	Compatibility with Digital Twins	Performance Metrics
Predictive analytics, demand planning, inventory optimization	Optimize inventory levels, reduces lead times, enhances forecasting accuracy	High: integration with DTs for real-time simulation and optimization	Demonstrates high reliability in demand forecasting
AI-driven risk management, real-time supply chain visibility	Identify potential risks, streamline logistics, improves service levels	Moderate: supports real-time insights but with limited DT-specific features	Consistently identifies and mitigates supply chain risks
AI and machine learning for disruption prediction, decision support	Provides early warnings for disruptions, supports recovery strategies	High: seamlessly integrates with DT platforms for resilience modeling	Recognized for accurate disruption prediction
End-to-end supply chain management, AI for demand forecasting	Improves demand planning accuracy, integrates supply chain operations	High: supports DT integration for comprehensive supply chain modeling	Highly effective in refining demand planning
IoT integration, AI-powered supply chain insights	Enhances visibility into operations, reduces downtime with predictive insights	Moderate: relies on IoT for real-time data sharing with limited DT integration	Effective in improving operational visibility and reducing downtime
IoT analytics, predictive maintenance and digital thread management	Monitors industrial equipment for predictive maintenance, reduces downtime, optimizes manufacturing processes	High: advanced DT integration for industrial applications	Strong performance in operational efficiency metrics

4.6. Digital Twin Technologies and Their Role in Enabling Decentralized Supply Chain Collaboration

Decentralized collaboration in supply chain management represents a strategic approach that skillfully combines the benefits of centralized and decentralized systems. This balance bolsters resilience against disruptions while optimizing efficiency and agility. Businesses facing the choice between centralizing or decentralizing their supply networks are increasingly leveraging the strengths of both paradigms. Centralized supply chains excel in maintaining standardized processes and ensuring global visibility, while decentralized systems offer the flexibility that allows local sites to address challenges creatively and promptly.

To support this hybrid approach, advanced supply chain management software has become indispensable. These solutions integrate disparate enterprise resource planning (ERP) systems, fostering a unified data model that serves as a single source of truth. This integration enhances cross-functional collaboration, enabling supply chain stakeholders to combine global oversight with localized execution. The result is a next-generation supply chain framework characterized by resilience, agility and efficiency.

Digital twins (DTs) have the potential to revolutionize supply chain networks by enabling a collaborative and decentralized model that emphasizes flexibility, transparency and resilience. By creating virtual representations of physical assets, processes and systems, DTs can transform traditional supply chains into adaptive, intelligent networks capable of responding to dynamic market conditions. Their integration with emerging technologies,

such as IoT, blockchain and AI, provides new opportunities to address the key challenges in achieving smooth coordination and decentralization. The table below (Table 5) illustrates how DTs can be utilized to address essential aspects of decentralized supply chain networks and identifies the technologies necessary for their deployment. It also demonstrates how these emerging technologies resolve specific challenges in implementation, ensuring seamless integration, scalability and operational efficiency.

Table 5. Applications of digital twins in collaborative, decentralized supply chain networks.

Aspect	How Digital Twins Can Be Used	Key Technologies to Support Digital Twins	Overcoming Challenges with Supporting Technologies
Decentralization	DTs can simulate and coordinate localized decisions, empowering each node in a supply chain to operate autonomously.	Blockchain enables secure data sharing, while distributed ledger technology (DLT) ensures decentralized governance.	Address lack of trust and data inconsistency through blockchain-enabled transparency and real-time synchronization.
Collaboration	DTs can act as a shared platform where stakeholders collectively plan, simulate and optimize operations.	APIs, cloud-based collaboration platforms and smart contracts facilitate real-time communication and coordination.	Overcome fragmented communication by using APIs and middleware to integrate stakeholders' systems into a unified platform.
Resilience	DTs can model and predict disruptions, allowing supply chains to test and adapt strategies dynamically.	AI and machine learning power predictive analytics, while edge computing enhances response times for local decisions.	Improve response times by leveraging edge computing for real-time decision-making and AI for disruption simulations.
Interoperability	DTs can integrate and standardize data from diverse systems to ensure compatibility across platforms.	Middleware solutions, REST APIs and integration platforms enable seamless communication across systems.	Solve system incompatibility through middleware and standardized APIs that allow for seamless data sharing between platforms.
Data Security and Sharing	DTs can implement secure models for data flows, ensuring sensitive information is shared safely across stakeholders.	Blockchain-based access control, encryption techniques and zero-knowledge proofs protect data privacy and security.	Mitigate data breaches by using encryption and ensure trust in shared data with blockchain-based immutable records.
Sustainability	DTs can simulate environmental impacts of supply chain decisions, guiding participants toward sustainable practices.	IoT sensors provide environmental metrics, while AI optimizes resource usage to reduce waste and emissions.	Overcome inefficiencies in tracking environmental impact by combining IoT sensors and AI for sustainability analysis.
Scalability	DTs can handle growing data and complexity as supply chain networks expand, ensuring efficiency at scale.	Scalable cloud infrastructures and decentralized storage systems like IPFS support larger networks.	Address the challenges of scaling data volume with cloud computing and distributed storage solutions for resilience.
Forecasting	DTs can analyze historical and real-time data to predict disruptions and suggest actionable responses.	AI-driven analytics tools and quantum computing enhance prediction accuracy and prescriptive insights.	Manage unpredictability by using AI to model scenarios and quantum computing to handle complex simulation environments.
Cost Optimization	DTs can simulate cost-efficient scenarios, helping stakeholders identify and reduce inefficiencies.	ERP systems, process automation tools and AI-powered optimization algorithms aid cost management.	Reduce financial waste by deploying AI-driven optimization algorithms and ERP tools for real-time cost tracking.

As can be seen from the table above, interoperability is a key challenge in the implementation of digital twins for SCR. Seamless interoperability among digital twins and related technologies across global supply chains is necessary to leverage real-time data, forecast simulations, and drive large-scale decision-making. Effective integration allows these systems to respond efficiently to organizational disruptions, rapidly shifting market dynamics and external challenges such as geopolitical risk or natural disasters. In this way, quick, seamless, secure and reliable data and information exchange is achieved between computing devices (viz. devices being able to transfer data), between information systems (of organizations, infrastructures, logistics networks), and between devices and systems, with the aim of enhancing cooperation or cooptation of independent logistics parties or networks. Developing common data standards and exchange formats is crucial to ensure different systems can communicate effectively.

Few frameworks and standards have been developed to strengthen interoperability. Some papers introduced digital twin system interoperability framework, emphasizing system-centric design, model-based approaches, and digital threads to enable scalable and dynamic systems integration [83]. This framework offers foundational guidance for creating interconnected digital twin ecosystems capable of real-time collaboration and decision-making. ISO 23247 is the standard for manufacturing and supporting interoperability by providing reference architectures, data exchange formats, and communication protocols for integrating digital twins across heterogeneous systems [84]. This standard enables the integration of digital twins in multi-vendor environments by addressing relevant communication and data integration needs. Interoperability challenges extend beyond technical limitations to include organizational and human factors. Wagg et al. (2020) identify the “silo effect” as a significant barrier where isolated systems and data sources hinder unified decision-making [63]. Standardized data formats and exchange formats are key to addressing these challenges. Privacy is also a significant obstacle to interoperability as a federated learning to balance the need for data sharing with privacy protections was proposed, enabling secure collaboration without compromising sensitive information [85].

Lehmann et al. (2023) highlights the role of Multi-Agent Systems (MASs) in fostering intelligent collaboration between digital twins, allowing them to interact seamlessly within complex ecosystems [64]. Similarly, Ye et al. (2023) emphasizes the importance of distributed data management in cloud-based systems to enable data fusion, which enhances the integration of digital twins across global supply chains [36]. In fact, while digital twins are widely recognized for their potential to enhance resilience in supply chain management, the literature still lacks sufficient evidence-based research on the practical effectiveness of interoperability standards for digital twins in these systems.

5. Conclusions

The global adoption of digital twin technology presents opportunities and challenges, varying significantly across regions. In developed areas, such as North America and Europe, the primary barriers stem from the complexities of high-tech integration. The seamless connection between physical assets and their digital replicas requires advanced infrastructure, robust data systems and significant investment in both hardware and software [61]. The need for continuous updates, cyber-security measures and real-time synchronization adds further layers of complexity, making widespread adoption costly and technically demanding. On the other hand, developing regions often face more foundational challenges. Many needs help with infrastructure gaps, such as inconsistent internet connectivity, outdated systems and limited access to cutting-edge technologies [86]. These regions are often hindered by a need for more skilled labor and expertise in implementing and maintaining digital twin systems, alongside the financial constraints that make large-scale digital transformations

more challenging to achieve. Nevertheless, developing regions are increasingly recognizing the value of digital twins in enhancing efficiency, reducing costs and improving overall supply chain performance. With strategic investments and international collaborations, these regions are gradually working toward overcoming these barriers.

On the other hand, in regions with limited infrastructure, scaling digital twin (DT) adoption requires strategic approaches to balance innovation with resource constraints. Cloud computing provides an efficient solution by reducing reliance on on-premises infrastructure, offering scalable and cost-effective data storage and processing. Edge computing complements this by enabling localized data analysis, minimizing bandwidth requirements in areas with limited connectivity. A modular approach to DT implementation allows for incremental adoption, focusing initially on high-priority areas such as asset monitoring or predictive maintenance. Open-source tools and affordable IoT devices provide cost-effective alternatives to proprietary systems, making adoption more accessible. Public-private partnerships can further facilitate investment in infrastructure and training, ensuring local expertise is developed to sustain and optimize these systems. Efficient data management, through compression and prioritization, enables DT systems to function effectively despite bandwidth limitations. Sustainability is also critical; integrating energy-efficient IoT devices and renewable energy can reduce operational costs and environmental impact.

Combining these strategies, regions with limited infrastructure can effectively scale digital twin adoption. This approach ensures that the transformative potential of DTs is realized while addressing the unique challenges of resource-constrained environments, fostering resilient, adaptive and efficient supply chain networks.

A crucial consideration in scaling digital twin technology lies in balancing operational efficiency with achieving sustainability goals, particularly in industries like manufacturing and logistics, which have significant environmental impacts. In manufacturing, digital twins can optimize energy consumption and reduce material waste by identifying inefficiencies in production processes. Predictive maintenance ensures equipment operates efficiently, avoiding energy spikes caused by malfunctions. Similarly, in logistics, DTs help reduce emissions by optimizing delivery routes and vehicle performance. By simulating real-world scenarios, digital twins enable organizations to test environmentally friendly practices, such as integrating electric vehicles or switching to renewable energy sources, without disrupting operations. These capabilities empower industries to meet sustainability targets while maintaining productivity.

Regions worldwide are actively investing in the future of digital twin technology, with different areas focusing on diverse priorities. In Europe, digital twins are being utilized extensively for sustainability initiatives, particularly within industries aiming to reduce carbon footprints and improve resource management. European countries are leveraging digital twins to optimize energy consumption, streamline production processes and contribute to the EU's green initiatives [61]. Meanwhile, in Asia, particularly in China, Japan and South Korea, cities are leading the way in integrating digital twin technology into smart city frameworks. These regions use digital twins to enhance transport management, increase the urban transport network performance, and improve the overall quality of life in densely populated areas. By simulating various urban scenarios and using real-time data, Asian cities can address congestion, energy use and emergency responses more effectively, pushing the boundaries of what digital twin technology can achieve.

As the global digital twin landscape evolves, its potential to transform supply chain management (SCM) becomes increasingly apparent. In today's complex and rapidly changing business environment, supply chain resilience has become a critical concern for organizations worldwide. The emergence of digital twin technology offers a powerful

solution to these challenges by providing enhanced visibility and predictive capabilities that allow supply chain managers to anticipate and mitigate disruptions before they occur [86].

A key benefit of digital twins in SCM is it helps managers identify inefficiencies, optimize logistics and make data-driven decisions that improve overall performance by offering a complete, up-to-date view of the supply chain. Furthermore, digital twins allow for scenario-based simulations, enabling organizations to test various strategies and see how different factors—such as changes in demand, supplier disruptions, or logistical bottlenecks—might impact the supply chain. This proactive approach can significantly reduce downtime and losses, helping companies respond swiftly and effectively to unexpected challenges [3].

In addition to their operational benefits, digital twins significantly enhance supply chain resilience. Digital twins can predict potential disruptions based on historical data, current trends and external factors by leveraging advanced data analytics, machine learning and AI. For example, by analyzing weather patterns, geopolitical risks, or shifts in consumer behavior, digital twins can provide early warnings of potential disruptions and suggest alternative routes or suppliers to mitigate risk [79]. This predictive capability is invaluable in today's uncertain business environment, where quickly adapting to disruptions can make the difference between success and failure.

Moreover, digital twin technology facilitates collaboration and coordination across the supply chain. By providing real-time data and insights to all stakeholders, from suppliers to manufacturers to distributors, digital twins enable more seamless communication and coordination, ensuring that everyone is aligned and working towards the same goals. This enhanced visibility and collaboration improves operational efficiency and contributes to building a more resilient, agile supply chain capable of adapting to the ever-changing demands of the global market. By identifying vulnerabilities and simulating disruptions (such as supplier delays or demand fluctuations), digital twins allow businesses to implement proactive risk management strategies. Additionally, investments in digital twins must align with organizational contexts, including regulatory environments and market conditions.

Moving forward, regional disparities in digital twin adoption and technical and infrastructural challenges will need to be addressed to unlock the full potential of this transformative technology. Refinement and scaling of existing implementations in developed regions are likely to focus on leveraging these advancements to optimize performance and expand applications. Meanwhile, developing areas must focus on foundational investments and capacity building to fully embrace digital twin's capabilities. Bridging these gaps will play a critical role in shaping the future of supply chain management and cultivating resilience in this ever-digitizing world. For digital twins to reach their full potential in supply chain resilience, significant gaps in evidence-based research on the practical effectiveness of interoperability standards must be addressed. This includes developing robust frameworks, scalable solutions, and ensuring data standardization and seamless integration between different Industry 4.0 technologies to enhance their effectiveness in dynamic and complex environments.

The potential of digital twin technology is immense, reaching far beyond its current applications. With advancements in AI, blockchain and IoT, digital twins are set to achieve new levels of precision and provide deeper insights into complex operations. As regions work to overcome their unique challenges, digital twins are well positioned to drive transformative advancements in sustainability, operational efficiency and resilience across a diverse range of industries. Digital twins can become the basis of the future generation of supply chain management by promoting cooperation between industries and tailoring approaches to context-specific requirements. This evolution promises a more connected, efficient and sustainable future, where digital twins serve as a vital tool for addressing the demands of a rapidly changing global landscape shaping the next generation of supply

chain management solutions and fostering resilience in a world increasingly reliant on digital innovation.

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